Stability Derivation

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0.1 Equation of motion derivation

For an object with mass m and moment of inertia matrix I_p about a pivot which has velocity $\mathbf{v}_p$, where the displacement from the pivot to the center of mass is $\mathbf{r}_{\text{pivot}\to\text{com}}$, and the object is rotating with angular velocity ω about the pivot, the kinetic energy is

$$T = \frac{1}{2}m\mathbf{v}_p^2 + m\mathbf{v}_p \cdot (\ \times \mathbf{r}_{\text{pivot}\to\text{com}}) + \frac{1}{2}\ \cdot \mathrm{I}_p\ .$$

For our pendulum, we have $\omega = \dot{\theta}\hat{k}$, so the only important component of moment of inertia is $I = \mathrm{I}_p[3,3]$. We also get $\ \times \mathbf{r}_{\text{pivot}\to\text{com}} = \dot{\theta}l\,\hat{\theta}$ where l is the distance from the pivot to the center of mass. We construct the pivot so that its velocity only in the $\hat{j}$ direction, so let's call the pivot's vertical position d and velocity $\dot{d}$.

We therefore have

$$T = \frac{1}{2}m\dot{d}^2 + ml\dot{d}\dot{\theta}\sin\theta + \frac{1}{2}I\dot{\theta}^2.$$

The potential energy is

$$V = mgy = mg(d - l\cos\theta).$$

Plugging these into the Euler-Lagrange equation, we get

$$\ddot{\theta} + \frac{ml}{I}(g - \ddot{d})\sin\theta = 0.$$

0.2 Reducing variables

If $\ddot{d} = 0$, $\omega_0^2 = \frac{mgl}{I}$ would be the natural frequency of the pendulum. Let's use that and also define $\ddot{d}_g = \ddot{d}/g$, a unitless quantity representing effective pivot acceleration relative to gravity. This lets us rewrite the equation of motion in terms of only a few variables:

$$\ddot{\theta} + \omega_0^2(1 - \ddot{d}_g)\sin\theta = 0.$$

For the basic Kaptiza pendulum, we have the pivot simply oscillating with amplitude A and frequency ν: $d = A\cos(\nu t)$. Therefore

$$\ddot{d}_g = -\frac{A}{g}\nu^2\cos(\nu t)$$

is the relative acceleration. This hints towards defining an amplitude relative to gravity $A_g = A/g$, giving us the equation of motion

$$\ddot{\theta} + \omega_0^2 \left(1 + A_g \nu^2 \cos(\nu t)\right) \sin\theta = 0$$

with only three variables describing the experimental setup: ω_0, A_g, and ν. Also note that ω_0 is easily measurable by keeping the pivot still and A_g and ν are chosen by the experimenter.

0.3 Stability of the Kaptiza Pendulum

Let's split θ into two parts: the fast-moving part coming directly from the vibration of the pivot ξ and the rest which we'll call θ_0, so that

$$\theta = \theta_0 + \xi$$

and

$$\ddot{\theta} = \ddot{\theta}_0 + \ddot{\xi}.$$

We'll assume that the pivot's oscillation is significantly faster than the natural frequency: $\nu \gg \omega_0$. Let's also assume that the amplitude of the pivot's motion is much smaller than the distance from the pivot to the center of mass: $l \gg A$. This means that ξ will be only a tiny motion. Therefore, it is a reasonable approximation to replace $\sin\theta$ with its Taylor expansion about $\theta = \theta_0$.

$$\sin\theta \approx \sin\theta_0 + \xi\cos\theta_0$$

The equation of motion can now be written as

$$\ddot{\theta}_0 + \ddot{\xi} + \omega_0^2 \left(1 + A_g \nu^2 \cos(\nu t)\right) (\sin\theta_0 + \xi\cos\theta_0) = 0.$$

We can make some more approximations now. Since ξ is "fast" and θ_0 is "slow", we can say $\ddot{\xi} \gg \ddot{\theta}_0$. Even though A_g is assumed to be small, ν being squared means that $A_g\nu^2 \gg 1$. We also reuse the idea that ξ is small, $\xi \ll 1$. Putting these together, we can remove terms and the expression simplifies to

$$\ddot{\xi} + \omega_0^2 A_g \nu^2 \cos(\nu t)\sin\theta_0 = 0.$$

Integrating twice and assuming that θ_0 is roughly constant for the time scale of interest (namely one period of the pivot's oscillation, which is very short), we get an expression for the fast motion as a function of time

$$\xi = A_g \omega_0^2 \sin\theta_0 \cos\nu t.$$

Now that we know $\ddot{\xi}$ and ξ, we can substitute back into the expanded equation of motion to get an equation of motion for θ_0:

$$\ddot{\theta}_0 - A_g\omega_0^2\nu^2 \sin\theta_0\cos\nu t + \omega_0^2\left(1 + A_g\nu^2\cos\nu t\right)\left(\sin\theta_0 + A_g\omega_0^2\sin\theta_0\cos\theta_0\cos\nu t\right) = 0,$$

or, expanding:

$$\begin{aligned}
&\ddot{\theta}_0 - A_g\omega_0^2\nu^2\sin\theta_0\cos\nu t + \omega_0^2\sin\theta_0 \\
&\quad + A_g\omega_0^4\sin\theta_0\cos\theta_0\cos\nu t + A_g\omega_0^2\nu^2\sin\theta_0\cos\nu t \\
&\quad + A_g^2\omega_0^4\nu^2\sin\theta_0\cos\theta_0\cos^2\nu t \\
&\quad = 0.
\end{aligned}$$

The next approximation trick is to time-average this equation over a period of the pivot's oscillation. Computationally, this means integrating from some time t' to $t' + \frac{2\pi}{\nu}$ then dividing by the time difference $\frac{2\pi}{\nu}$. Since θ_0 moves very slowly relative to this time scale, we can say that θ_0 and $\ddot{\theta}_0$ are constant for this integration. There are two important identities to note:

$$\int_{t'}^{t'+\frac{2\pi}{\nu}} \cos(\nu t)dt = 0$$

and

$$\int_{t'}^{t'+\frac{2\pi}{\nu}} \cos^2(\nu t)dt = \frac{1}{2} \cdot \frac{2\pi}{\nu}.$$

So integrating the above expression, the result is

$$\ddot{\theta}_0 + \omega_0^2 \sin\theta_0 + \frac{1}{2}A_g^2\omega_0^4\nu^2 \sin\theta_0 \cos\theta_0 = 0,$$
$$\ddot{\theta}_0 + \omega_0^2 \left(1 + \frac{1}{2}A_g^2\omega_0^2\nu^2 \cos\theta_0\right) \sin\theta_0 = 0.$$

This is now an equation for the general motion of the pendulum, and notice it has no time dependence! We can check its stability in the inverted position $\theta_0 = \pi$. When $\theta_0 \approx \pi$, $\sin\theta_0 \approx -\theta_0$ and $\cos\theta_0 \approx -1$. Therefore,

$$\ddot{\theta}_0 - \omega_0^2(1 - \frac{1}{2}A_g^2\omega_0^2\nu^2)\theta_0 = 0$$

which means the pendulum is stable when

$$-\omega_0^2(1 - \frac{1}{2}A_g^2\omega_0^2\nu^2) > 0$$
$$\implies \omega_0^2\nu^2A_g^2 > 2.$$

0.4 Stability of Kaptiza's pendulum driven by any periodic function

As found before, the general equation of motion for a pendulum with an accelerating pivot is

$$\ddot{\theta} + \omega_0^2(1 - \ddot{d}_g)\sin\theta = 0.$$

Previously, we said that the pivot is oscillating sinusoidally, so $\ddot{d}_g = -A_g\nu^2 \cos\nu t$. Let's instead drive the pivot by a more general function made up of sinusoids:

$$d_g = \sum_{n=0}^{\infty} A_n \sin(n\nu_1 t) + B_n \cos(n\nu_1 t).$$

This is a Fourier series, and therefore it can represent any periodic function. For brevity, I'll write $\nu_n = n\nu_1$ and $\mathbf{d}_n = A_n \sin(\nu_n t) + B_n \cos(\nu_n t)$. We can write the second derivative as

$$\ddot{d}_g = -\sum_{n=0}^{\infty} \nu_n^2 \mathbf{d}_n.$$

Substituting into the equation of motion:

$$\ddot{\theta} + \omega_0^2(1 + \sum_{n=0}^{\infty} \nu_n^2 \mathbf{d}_n)\sin\theta = 0.$$

We can then do the exact same ξ trick as before, using the same assumptions. Namely, we must have that $\sum_{n=0}^{\infty} \nu_n^2 \mathbf{d}_n \ll 1$. Also, even the lowest frequency $\nu_1 \gg \omega_0$. We end up with a similar long expression for θ_0:

$$\begin{aligned}
&\ddot{\theta}_0 - \omega_0^2 \left(\sum_{n=0}^{\infty} \nu_n^2 \mathbf{d}_n \right) \sin\theta_0 \\
&\quad + \omega_0^2 \sin\theta_0 + \omega_0^4 \left(\sum_{n=0}^{\infty} \mathbf{d}_n \right) \sin\theta_0 \cos\theta_0 \\
&\quad + \omega_0^2 \left(\sum_{n=0}^{\infty} \nu_n^2 \mathbf{d}_n \right) \sin\theta_0 \\
&\quad + \omega_0^4 \left(\sum_{n=0}^{\infty} \nu_n^2 \mathbf{d}_n \right) \left(\sum_{m=0}^{\infty} \mathbf{d}_m \right) \sin\theta_0 \cos\theta_0 \\
&\quad = 0.
\end{aligned}$$

We can how do the same averaging trick. To get convenient results from the integrals, we must integrate across a period which is an integer multiple of the period of each sinusoid in the sum. This is where it becomes important that d_g is a Fourier *series*. There exists such a period, and it's $T = \frac{2\pi}{\nu_1}$. For sinusoid n with frequency $\nu_n = n\nu_1$, the period is $T_n = \frac{2\pi}{n\nu_1}$. Multiplying by integer n, notice we get $nT_n = T$. We can use some nice observations for the integrals:

$$\begin{aligned}
\int_{t'}^{t'+T} \sum_{n=0}^{\infty} \nu_n^2 \mathbf{d}_n dt &= \sum_{n=0}^{\infty} \nu_n^2 \int_{t'}^{t'+T} \mathbf{d}_n dt \\
&= \sum_{n=0}^{\infty} \nu_n^2 \left(A_n \int_{t'}^{t'+T} \sin(\nu_n t) dt + B_n \int_{t'}^{t'+T} \cos(\nu_n t) dt \right) \\
&= 0.
\end{aligned}$$

And for the last term,

$$\begin{aligned}
\int_{t'}^{t'+T} \left(\sum_{n=0}^{\infty} \nu_n^2 \mathbf{d}_n \right) \left(\sum_{m=0}^{\infty} \mathbf{d}_m \right) dt &= \int_{t'}^{t'+T} \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \nu_n^2 \mathbf{d}_n \mathbf{d}_m dt \\
&= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \nu_n^2 \left(\int_{t'}^{t'+T} \mathbf{d}_n \mathbf{d}_m dt \right)
\end{aligned}$$

We can greatly simplify that integral:

$$\begin{aligned}
\int_{t'}^{t'+T} \mathbf{d}_n \mathbf{d}_m dt &= A_n A_m \int_{t'}^{t'+T} \sin\nu_n t \sin\nu_m t dt \\
&\quad + A_n B_m \int_{t'}^{t'+T} \sin\nu_n t \cos\nu_m t dt \\
&\quad + B_n A_m \int_{t'}^{t'+T} \cos\nu_n t \sin\nu_m t dt \\
&\quad + B_n B_m \int_{t'}^{t'+T} \cos\nu_n t \cos\nu_m t dt \\
&= A_n A_m \frac{T}{2} \delta_{n,m} + 0 + 0 + B_n B_m \frac{T}{2} \delta_{n,m}
\end{aligned}$$

$$\implies \int_{t'}^{t'+T} \left(\sum_{n=0}^{\infty} \nu_n^2 \mathbf{d}_n\right)\left(\sum_{m=0}^{\infty} \mathbf{d}_m\right) dt = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} \nu_n^2 \left(A_n A_m \frac{T}{2}\delta_{n,m} + B_n B_m \frac{T}{2}\delta_{n,m}\right)$$
$$= \frac{T}{2}\sum_{n=0}^{\infty} \nu_n^2 \left(A_n^2 + B_n^2\right)$$

We can define $R_n^2 = A_n^2 + B_n^2$ to be the amplitude of the $\mathbf{d}_n$ term. Putting everything back into the θ_0 equation:

$$\ddot{\theta}_0 + \omega_0^2 \left(1 + \frac{1}{2}\omega_0^2 \cos\theta_0 \sum_{n=0}^{\infty} \nu_n^2 R_n^2\right) \sin\theta_0 = 0.$$

Similarly to before, we get stability if

$$\omega_0^2 \sum_{n=0}^{\infty} \nu_n^2 R_n^2 > 2.$$

0.4.1 Parseval's Theorem

Parseval's theorem states that, for a series like this,

$$\frac{1}{T}\int_0^T [x(t)]^2 dt = \frac{B_0^2}{4} + \frac{1}{2}\sum_{n=1}^{\infty} R_n^2.$$

Now let $x' = \frac{dx}{dt}$, and notice:

$$\frac{d}{dt}\left(\sum_{n=0}^{\infty} A_n \sin(\nu_n t) + B_n \cos(\nu_n t)\right) = \sum_{n=1}^{\infty} A_n \nu_n \cos(\nu_n t) - B_n \nu_n \sin(\nu_n t).$$

Applying Parseval's theorem to x', where $R_n'^2 = A_n^2\nu_n^2 + B_n^2\nu_n^2 = \nu_n^2 R_n^2$,

$$\frac{1}{T}\int_0^T [x'(t)]^2 dt = 0 + \frac{1}{2}\sum_{n=1}^{\infty} \nu_n^2 R_n^2$$
$$\implies \sum_{n=0}^{\infty} \nu_n^2 R_n^2 = \frac{2}{T}\int_0^T [x'(t)]^2 dt.$$

Notice that $\sum_{n=1}^{\infty} \nu_n^2 R_n^2 = \sum_{n=0}^{\infty} \nu_n^2 R_n^2$ since the $n = 0$ term is 0 anyway. This is an easier way to calculate the bound for any periodic function with period T!

$$\omega_0^2 \int_0^T (\dot{d}_g)^2 dt > T$$

0.4.2 Example 1: Triangle waves

We know that the frequencies at which a sine wave makes a stable pendulum is

$$\nu > \frac{\sqrt{2}}{\omega_0 A_g}.$$

What about at triangle wave?

A triangle wave with peak value A_g and frequency ν has the Fourier series representation

$$d_g = \frac{8A_g}{\pi^2} \sum_{n=1,3,5,\ldots}^{\infty} \frac{(-1)^{(n-1)/2}}{n^2} \cos(n\nu t)$$

so we have $A_n = 0$, $B_n^2 = \left(\frac{8A_g}{\pi^2 n^2}\right)^2$ and therefore

$$\begin{aligned}
\sum_{n=0}^{\infty} \nu_n^2 R_n^2 &= \sum_{n=1,3,5,\ldots}^{\infty} n^2\nu^2 \frac{64A_g^2}{\pi^4 n^4} \\
&= \frac{64A_g^2\nu^2}{\pi^4} \sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{n^2} \\
&= \frac{64A_g^2\nu^2}{\pi^4} \cdot \frac{\pi^2}{8} \\
&= \frac{8A_g^2\nu^2}{\pi^2}
\end{aligned}$$

and so the stability condition is

$$\begin{aligned}
\omega_0^2 \frac{8A_g^2\nu^2}{\pi^2} &> 2 \\
\nu &> \frac{\pi}{2\omega_0 A_g}.
\end{aligned}$$

Note that $\pi/2$ is slightly greater than $\sqrt{2}$. The Kapitza pendulum should be slightly more stable when the pivot is oscillating with a sine wave than with a triangle wave!

0.4.3 Example 2: Square waves

Square waves actually give an inconclusive result! The series does not converge. Maybe this is expected, square waves have discontinuities when jumping from low to high, so the acceleration briefly jumps up to ∞. So let's smooth out the jumps a bit with a trapezoidal wave:

(work in progress)